[15] Comms - EtherCat

H1102NL magnetics, Bob Smith (75R + 1n 2kV), JST-GH connectors

ETHERCAT — LAN9253 (SPI PDI, 2 PORTS), Magnetics, SII EEPROM

49.9R terminations to +3V3_PHY (at the LAN9253 pins)

U9 — LAN9253-V/R4X

Mode strap, reset, SPI MISO series

25 MHz crystal

SII EEPROM (I2C address 0x50)

EtherCAT indicators + strap resistors (straps read at reset)

+3V3: VDD33 (5) 1u+100n, VDDIO (14, 20, 32, 37, 47) 100n each, U3 100n

+3V3_PHY: pins 51, 58, 64, T1/T2 centre taps, FB5 output

+1V2_ESC (internal regulator): C161 1u low-ESR (DS00003421B 4.1.1) at VDDCR pin 6; 24, 38, 56, 59 100n each

DESIGN NOTE:
WP low: master can write the SII

LAYOUT NOTE: ETHERCAT
* TX/RX pairs 100 ohm differential, matched in-pair, short; 49.9R at the U9 pins.
* Keep PHY side and cable side of T1/T2 apart; no GND plane under the cable side and the GH connectors except the Bob Smith cap return; C155/C156 right at the centre taps, returned to CHASSIS.
* Y1 + caps and R55 (RBIAS) tight to their pins; exposed pad via array to GND.

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